

Business Analytics, 2e

APRIORI

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Chapter 14 Learning Objectives (LOs)

LO 1 Concept of Association rules mining

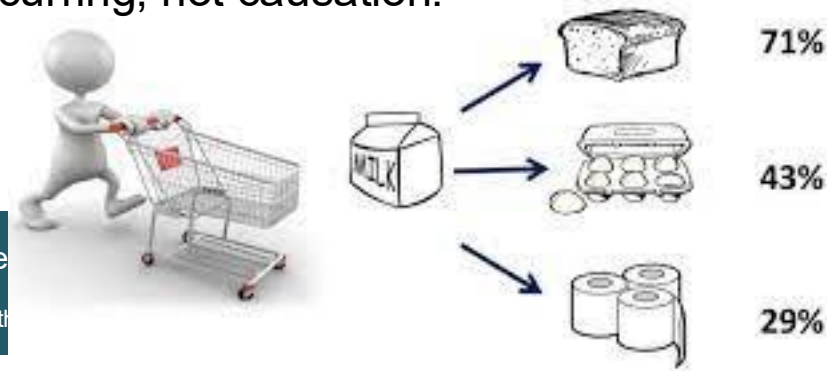
LO 2 Apriori algorithm

LO 3 Measure the effectiveness of Association rules

- Support
- Confidence
- Lift
- Conviction

Association Rule Analysis

- Association rule analysis is essentially a “**what goes with what**” study.
 - Designed to identify events that tend to **occur together**
 - Also known as *affinity analysis* or *market basket analysis*
- Association rules are **If-Then logical statements** that represent relationships among different items, usually called itemset.
 - Item: A single product or entity
 - "Milk", "Bread"
 - Itemset: A set of items that appear together in a transaction
 - {"Milk", "Bread"}, {"Cookies", "Butter"}
- Designed to *identify hidden patterns* and co-occurring events in data
 - “**If**” is the **antecedent**,
 - “**then**” is the **consequent**
 - If-Then in the sense of frequency of co-occurring, not causation.



Use cases in various industries

- Explain patterns in data from various information repositories, such as relational databases and transactional databases.

User experience (UX) design.

- Developers can collect data on how consumers use a website they create.
 - Use associations in the data to optimize the website user interface
 - Where users tend to click and what maximizes the chance that they engage with a call to action

Entertainment.

- Services like Netflix and Spotify can use association rules to fuel their content recommendation engines.
 - Analyze past user behavior data for frequent patterns to:
 - Recommend content that a user is likely to engage with, or,
 - Organize content in a way that is likely to put the most interesting content for a given user first.

Sample Case: UX/UI Design by Clickstream sessions

- Define buttons on pages
 - Usually **Target event** = CallToAction click (e.g., sign up, checkout)
 - See the ClickStream.ipynb

```
sessions = [  
    ["Home", "About", "Contact", "CallToAction"],  
    ["Home", "Features", "Pricing", "CallToAction"],  
    ["Home", "Features", "Exit"],  
    ["Home", "Blog", "Newsletter"],  
    ["Home", "Pricing", "CallToAction"],  
    ["Home", "Blog", "Features", "CallToAction"],  
    ["Home", "About", "Exit"],  
    ["Home", "Blog", "Newsletter", "CallToAction"],  
    ["Home", "Features", "Pricing", "Contact", "CallToAction"],  
    ["Home", "Features", "Exit"],  
]
```

Association Rule Analysis

- Let n be the number of items.
- The number of possible combinations exponentially increases:
 - Often there are thousands or tens of thousands of cases
 - The search problem becomes extremely computationally intensive and time-consuming
- There are several algorithms that can be used to perform association rule analysis in a more efficient manner.
 - They all focus on the frequency of itemsets.
 - One of the most widely used methods is called the [Apriori algorithm](#).
 - [Others are, for example:](#) Eclat algorithm, FP-growth algorithm, etc.

Association Rule Analysis

- The Apriori algorithm is designed to recursively generate itemsets that exceed a predetermined frequency threshold.
- The term “support” of the item or itemset is often used synonymously with the frequency of the item or itemset.
 - Set a minimum support value
 - Below which an item or itemset is excluded
 - This makes the analysis more computationally feasible
- Eliminates infrequent items that are below the support value
- *This makes it easier to analyze relevant information in a large data set.*

Association Rule Analysis

- With enough data, we can propose many of these **If-Then association rules**.
- We need a way to **evaluate the effectiveness** of these rules, the strength of associations among products.
- Only the strong associations that occur frequently have the potential to reappear consistently in the future.
- **Support** of the association rule is the probability of the If-Then statement.

$$\frac{\text{Number of transactions including both antecedent and consequent}}{\text{Total number of transactions}}$$

- **Confidence** of the association rule is the conditional probability that the consequent occurring given the antecedent occurs.

$$\frac{\text{Number of transactions including both antecedent and consequent}}{\text{Number of transactions including antecedent}}$$

Look for an association rule that has high support and confidence.

Association Rule Mining

A Generic Rule: $X \Rightarrow Y$ [S%, C%]

X, Y: products and/or services

X: Left-hand-side (LHS) = Antecedent

Y: Right-hand-side (RHS) = Consequent

S: Support: how often **X** and **Y** go together

C: Confidence: how often X and Y go together, given that X goes

Example:

{Laptop Computer, Antivirus Software} $\Rightarrow$ {Extended Service Plan} [S=30%, C= 70%]

Association Rule Analysis

- In some cases, the antecedent and consequent are common (high Support and Confidence) yet unrelated.
 - There is not necessarily a relationship.
- The “**Lift**” ratio evaluates the strength of the association by
 - How much more often X and Y occur together than expected if they were independent.

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X) \times \text{Support}(Y)}$$

Lift	Statistical Meaning	Effect of X on Y	Business Insight Example
> 1	Positive correlation	X increases chance of Y	Buyers of coffee also tend to buy muffins
= 1	Statistical independence	X has no effect on Y	Buying bread doesn't affect buying eggs
< 1	Negative correlation	X reduces chance of Y	Buyers of soda are less likely to buy water

Support, Confidence and Lift calculation

$$Supp(X \rightarrow Y) = \frac{\text{no.of instances that contains both } X \text{ and } Y}{\text{total number of instances}}$$

$$Conf(X \rightarrow Y) = \frac{Supp(X, Y)}{Supp(X)}$$

$$Lift(X \rightarrow Y) = \frac{Supp(X, Y)}{Supp(X) * Supp(Y)}$$

Association Rule Mining– Apriori algorithm

1. Generate frequent itemsets of length 1 ($k=1$)
 - Any items with support $<$ threshold are pruned (eliminated)
2. Generate itemsets of length $k+1$
 - Prune (eliminate) candidate itemsets with support $<$ threshold value (non-frequent items)
3. Repeat 2 until no itemsets at that length or at the max k we define.
4. Generate association rules ($X \rightarrow Y$), ensure the confidence is as required (confidence = % the rules happened out of the antecedent (X)).
 - # e.g. if Conf = 0.8, $\{X\}$ of 100 occurrences, $\{X,Y\}$ happened 80 times.
5. Compute Lift to ensure the strength of the association. (Ensure Lift $>$ 1)

Association Rule Mining

Raw Transaction Data

Transaction No	SKUs (Item No)
1	1, 2, 3, 4
1	2, 3, 4
1	2, 3
1	1, 2, 4
1	1, 2, 3, 4
1	2, 4

One-item Itemsets

Itemset (SKUs)	Support
1	3
2	6
3	4
4	5

Two-item Itemsets

Itemset (SKUs)	Support
1, 2	3
1, 3	2
1, 4	3
2, 3	4
2, 4	5
3, 4	3

Three-item Itemsets

Itemset (SKUs)	Support
1, 2, 4	3
2, 3, 4	3

Assume required support = 2 / 6

Rule of 1--> 2: Supp. = 3/6, Conf. = (3/6)/(3/6), Lift = (3/6)/[(3/6)*(6/6)]

2 --> 3: Supp. = 4/6, Conf. = (4/6)/(6/6), Lift = (4/6)/[(6/6)*(4/6)]

1,2 --> 4: Supp. = 3/6, Conf. = (3/6)/(3/6), Lift = (3/6)/[(3/6)*(5/6)]

Association Rule Analysis

- Example: Consider 10 retail transactions of cosmetics.

Transaction	Items
001	lipstick, mascara, eye liner
002	eye shadow, mascara
003	primer, concealer, lip gloss
004	face brush, concealer, primer
005	mascara, lipstick, eye liner
006	mascara, eye liner, eye shadow
007	lip gloss, eye liner, mascara
008	mascara, eye shadow, lip gloss
009	face brush, mascara, primer, eye liner
010	eye liner, lipstick, primer

- Construct a frequency distribution that summarizes the number of purchases for each cosmetic product. Based on the frequency distribution, develop an association rule between the two most frequently purchased products.
- Compute and interpret the support, the confidence, and the lift ratio of the association rule developed in part a.

Association Rule Analysis

Mascara	Eye liner	Primer	Eye shadow	Lip gloss	Lipstick	Concealer	Face brush
7	6	4	3	3	3	2	2

- For $X=\text{Mascara}$, $Y=\text{Eye liner}$*

$\text{Support}(X,Y) = \frac{5}{10} = 0.50$

$\text{Conf}(X,Y) = \frac{5}{7} = 0.71$

$\text{Lift}(X,Y) = \frac{\text{Conf}(X,Y)}{\text{Support}(Y)} = \frac{0.71}{0.60} = 1.19$

$\text{Conviction} = \frac{1-\text{Support}(y)}{1-\text{Conf}(X,Y)} = \frac{1-0.6}{1-0.71} = 1.4$

For $X=\text{Eye liner}$, $Y=\text{Mascara}$

$\frac{5}{10} = 0.50$

$\frac{5}{6} = 0.83$

$\frac{0.83}{0.70} = 1.19$

$\frac{1-0.7}{1-0.83} = 1.8$
- The Lift > 1, indicate a strong association between the purchase of mascara and eyeliner.
 - The association is 19% stronger than guessing at random.
 - Note that Lift is Symmetric $\rightarrow \text{Lift}(X,Y) = \text{Lift}(Y,X)$
- The Conviction says Eye liner $\rightarrow$ Mascara is stronger the opposite

Feature	Lift	Conviction
Primary Goal	Measures strength of association (positive or negative) between items.	Measures the strength of <i>implication</i> in a directional rule (how much X implies Y).
Formula	$\frac{Support(X \cup Y)}{Support(X) \times Support(Y)}$	$\frac{1 - Support(Y)}{1 - Confidence(X \rightarrow Y)}$
Measures Against	Independence of X and Y.	Independence of X and $\neg Y$ (how often rule $X \rightarrow Y$ is <i>incorrect</i> vs. random chance).
Value = 1	X and Y are statistically independent.	Rule $X \rightarrow Y$ is correct only by chance (Confidence = Support(Y)). X and $\neg Y$ are independent.
Value > 1	Positive association. X and Y occur together more than expected.	Positive implication. Rule $X \rightarrow Y$ is correct more often than chance. High value = strong rule.
Value < 1	Negative association. X and Y occur together less than expected (substitutes).	Indicates negative correlation (rule $X \rightarrow Y$ is correct <i>less</i> often than chance).

Lift vs Conviction

Use **Lift** to find items that co-occur significantly more or less often than expected by chance.

It's great for identifying both **complementary items** ($\text{Lift} > 1$) and potential **substitutes** ($\text{Lift} < 1$).

Use **Conviction** to assess how strongly the presence of X implies the presence of Y in a directional rule.

It's useful for evaluating rules you might want to act upon, as high conviction suggests the rule reliably holds true much more than random chance.

Association Rule Analysis

- Example: The store manager at an electronics store collects data on the last 100 transactions.
- Five possible products were purchased: a keyboard, an SD card, a mouse, a USB drive, and/or a headphone.

Transaction 1	keyboard, mouse, headphone
Transaction 2	keyboard, SD card, mouse
⋮	⋮
Transaction 100	mouse, headphone

- Conduct association rule analysis to help the store manager understand the relationships among these products.

Conclusion

- Association rule is an unsupervised machine learning which can be used for data mining.
- It can help to support prescriptive analysis for business analytic where the recommendation or suggestions can be obtained with confidence.